

Data Structure and Algorithm LAB PROJECT

Lab Instructor: Ms. Rabeeya Saleem

Session: 2024 (Fall 2025)

A **Social Media Feed System** is modern, high-scoring, easy to explain, and beautifully integrates multiple data structures like queues, stacks, linked lists, arrays, and sorting/searching algorithms.

Social media platforms such as Twitter, Facebook, and Instagram rely heavily on data structures to efficiently handle post feeds, user interactions, and timelines. Students often don't see how these real systems internally manage posts, search content, handle recent activity, or maintain undo/redo operations.

This project implements a simplified but realistic social media feed using fundamental data structures to demonstrate how such platforms work under the hood.

How the Project Works

This system simulates a mini social media platform where users can:

- Add posts
- Delete posts
- View feed in FIFO manner
- Like posts
- Search posts
- Sort posts
- Navigate feed with "Back" using a stack
- Maintain friend lists using linked lists

Detailed System Modules

Module 1: User Management

- Create account
- View profile
- Edit profile
- Maintain friends list (linked list)

Module 2: Post Management

- Create post (enqueue)
- Edit post
- Delete post
- Like/unlike post
- Comment on post

Module 3: Feed Management (Core DS Logic)

Handles how the feed is built using:

- Queue for new posts

- Sorting for trending posts
- Searching for filtering posts

Module 4: Browsing History Module

- Tracks user navigation with stack
- Supports Undo:
 - Undo view
 - Undo like
 - Undo post delete

Module 5: Search & Sorting Module

- Keyword search
- Sort feed
- Sort friends list
- Search friend

Expected Outcomes

- Appropriate Name to Social Media Application
- A document that provides details about where and why did you use the specific data structure
- A Running Application with Text-File Database.

Programming Language: C++ / Java /C# (Provide Proper UI/UX)

Document Guidelines:

- Project Title
- Objectives
- Data Structure Used
- System Flow Diagram
- Implementation Details
- Case Study (Example Test Case)
- Screenshots of Working Output
- Performance Analysis (Time and Space Complexity)

Submission Guideline:

Each student require to work separately on this project. A zip folder containing implemented project with database and source code files. A single word document that should cover the above mentioned items.

Deadline for submission of this project till 24th November, 2025 10:00PM.